

A Direct JWST/NIRCam F150W Surface Brightness Fluctuation Calibration from the GO-3055 TRGB Sample

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Abstract

We present a direct empirical calibration of surface brightness fluctuation (SBF) magnitudes measured with JWST/NIRCam in F150W. The sample comprises the 14 early-type galaxies observed in JWST program GO-3055 for the TRGB–SBF distance-ladder project. We follow the analysis geometry of Jensen et al. (2025, arXiv:2502.15935), measuring SBF and color in two concentric circular annuli spanning 8.2–16.4 and 16.4–32.8 arcsec and combining the two measurements with inverse-variance weights. F090W–F150W colors are measured in the same annuli as the SBF signal. Absolute fluctuation magnitudes are anchored by the revised, color-dependent JWST TRGB distances of Chazov et al. (2026, arXiv:2603.11160). For the full sample we obtain the preliminary relation

$$\overline{M}_{150} = (-3.173 \pm 0.023) + (0.666^{+0.423}_{-0.476})[C_{090-150} - 0.5907],$$

with an inferred intrinsic scatter of 0.069 mag. Leave-one-out cross-validation against TRGB gives a mean residual of +0.008 mag and an RMS of 0.099 mag, corresponding to approximately 4.7 per cent in distance. A color-independent model is statistically competitive ($\Delta\text{AICc} = 1.71$ relative to the linear relation), so the present sample does not yet provide decisive evidence for the color slope. Measurements in the inner and outer annuli show materially different trends, including a highly significant 0.168 mag radial difference in NGC 3379 despite an exceptionally uniform residual image. We discuss the implications for stellar-population corrections and for transferring the TRGB zero point to future JWST SBF samples.

1 Introduction

Surface brightness fluctuations arise from Poisson variations in the number and luminosity of unresolved stars within each resolution element [1]. For old stellar populations the near-infrared SBF signal is strong, allowing efficient distance measurements to early-type galaxies well beyond the reach of resolved-star methods. Its absolute amplitude, however, depends on the luminosity function of the evolved stellar population and therefore must be calibrated as a function of a population proxy, normally an integrated optical or near-infrared color.

The TRGB–SBF project is constructing a Population-II distance ladder in which a geometrically calibrated tip of the red giant branch (TRGB) zero point is transferred to SBF measurements. Jensen et al. [4] (hereafter Paper III) used JWST TRGB distances for the GO-3055 galaxies to recalibrate the existing HST/WFC3-IR F110W SBF scale. Their analysis retained the established HST SBF measurements and used DECam or Pan-STARRS $g - z$ colors. With cluster-mean TRGB distances for Virgo and Fornax, they obtained

$$\overline{M}_{110} = (1.86 \pm 0.16)[(g - z)_{\text{PS}} - 1.30] - (2.760 \pm 0.016).$$

The revised calibration produced $H_0 = 73.8 \pm 0.7$ (stat) ± 2.3 (sys) $\text{km s}^{-1} \text{Mpc}^{-1}$.

Table 1: GO-3055 F150W SBF calibration measurements. The final column is the leave-one-out SBF distance residual relative to the adopted Paper IV TRGB modulus. Quoted values are preliminary and exclude a complete systematic error budget.

Galaxy	Environment	$C_{090-150}$ (mag)	$\overline{m}_{150}$ (mag)	$\overline{M}_{150}$ (mag)	$ \Delta\overline{m} _{\text{rings}}$ (mag)	μ_{TRGB} (mag)	$\Delta\mu_{\text{LOO}}$ (mag)
NGC 1380	Fornax	0.522	28.120	-3.288	0.046	31.408	-0.100
NGC 1399	Fornax	0.578	28.186	-3.312	0.019	31.498	-0.143
NGC 1404	Fornax	0.567	28.212	-3.154	0.026	31.366	+0.039
NGC 1549	Other	0.541	27.992	-3.239	0.046	31.231	-0.043
NGC 3379	Other	0.575	27.055	-3.023	0.168	30.078	+0.168
NGC 4374	Virgo	0.594	27.985	-3.127	0.043	31.112	+0.047
NGC 4406	Virgo	0.637	27.878	-3.218	0.062	31.096	-0.089
NGC 4472	Virgo	0.640	27.775	-3.241	0.039	31.016	-0.122
NGC 4486	Virgo	0.668	27.890	-3.156	0.069	31.046	-0.050
NGC 4552	Virgo	0.598	27.897	-3.026	0.086	30.923	+0.153
NGC 4621	Virgo	0.588	27.734	-3.087	0.072	30.821	+0.095
NGC 4636	Virgo	0.625	27.941	-3.129	0.022	31.070	+0.023
NGC 4649	Virgo	0.641	27.974	-3.035	0.119	31.009	+0.118
NGC 4697	Virgo	0.533	27.124	-3.200	0.114	30.324	+0.013

The present study takes the next direct step: both the fluctuation signal and the population color are measured from the JWST GO-3055 images themselves. F150W provides the SBF signal and F090W–F150W the color. The TRGB anchors are the thin-slice distance moduli in Chazov et al. [5] (hereafter Paper IV), who calibrated the color dependence of the F090W TRGB using the NGC 4258 maser distance. Paper IV found $M_{\text{F090W}}^{\text{TRGB}} = -4.398 \pm 0.017$ mag for $(F090W - F150W)_0 < 1.65$ mag, followed by a polynomial decline at redder color. Their thin-slice distances are on average 0.028 ± 0.008 mag shorter than the earlier literature values derived from the same imaging. That revision must be kept explicit when comparing our F150W zero point with Paper III.

Our goals are to (1) reproduce the Paper III annular measurement geometry, (2) establish a direct JWST F150W SBF calibration, (3) cross-validate SBF distances against the independent TRGB measurements, and (4) test whether radial population variations and host-galaxy luminosity explain part of the remaining scatter.

2 Data

2.1 GO-3055 sample

The sample contains 14 galaxies: three in the Fornax cluster, nine in the Virgo region, and two in nearby groups. Each target has NIRCcam F090W and F150W imaging. F150W is used for the fluctuation measurement and F090W is combined with it to form the population-sensitive color $C_{090-150} = F090W - F150W$. The adopted TRGB distance moduli are the thin-slice values from Paper IV. These distances are tied to the NGC 4258 maser modulus of 29.397 mag.

Table 2: Relationship of the present analysis to Papers III and IV.

Element	Paper III: Jensen et al. (2025)	Paper IV: Chazov et al. (2026)	Present work
Primary quantity	HST/WFC3-IR F110W SBF	JWST/NIRCam F090W TRGB	JWST/NIRCam F150W SBF
Population color	DECam or Pan-STARRS $g - z$	Resolved-star F090W–F150W	Integrated F090W–F150W in the SBF annuli
Distance anchor	JWST TRGB; mainly Virgo/Fornax cluster means	NGC 4258 maser; individual thin/thick-slice TRGB distances	Individual Paper IV thin-slice TRGB distances
Radial regions	8.2–16.4 and 16.4–32.8 arc-sec; weighted mean	Outer-halo resolved-star fields	Same two SBF annuli as Paper III; weighted mean and ring-by-ring tests
Color fit	Bayesian fit accounting for both-axis uncertainties	Piecewise TRGB color law with a break at 1.65 mag	Likelihood fit with intrinsic scatter; color error disabled in the current default fit
Calibration size	14 cluster-calibrated objects; eight direct TRGB overlaps	16 distance targets, 12 high-quality color-relation galaxies	14 direct SBF–TRGB pairs

Table 3: Paper IV TRGB quantities relevant to the present SBF calibration.

Quantity	Paper IV value
Blue TRGB zero point	-4.398 ± 0.017 mag
Color break	$F090W - F150W = 1.65$ mag
Red-branch coefficients (A, B, C, D)	$(-4.398, 0.431, 0.712, -0.293)$
Thin minus literature modulus	-0.028 ± 0.008 mag
Thick minus literature modulus	-0.037 ± 0.009 mag
Mean thin–thick difference	0.007 mag
Common zero-point systematic	0.047 mag
Choice adopted here	thin-slice moduli

2.2 Direct comparison with Papers III and IV

Table 2 separates differences forced by the instrument and filter set from genuine methodological choices. The most important choice still requiring a submission-level sensitivity test is the use of individual Paper IV TRGB distances here, whereas Paper III adopted cluster means for most Virgo and Fornax calibrators to reduce scatter and increase the color baseline.

3 SBF analysis

3.1 Galaxy model, masks, and residual image

A scalar sky level is estimated outside the science annuli and subtracted from the F150W image. A first mask removes invalid pixels, detector gaps, foreground objects, and compact sources. Isophotes are fitted to the usable galaxy light and interpolated to form a full-frame smooth model. Detector gaps are filled only for the purpose of obtaining a stable isophotal solution; they do not contribute to the SBF

measurement.

The smooth model is subtracted from the background-corrected image. Compact sources detected in the residual image are then masked, and the residual is normalized by the square root of the smooth galaxy model. A symmetric 3.5σ clipping of extreme residual pixels is retained as a deliberate modification to the historical pipeline. Its effect on the final zero point must be reported as a sensitivity test rather than silently absorbed into the method.

3.2 Power spectrum and unresolved sources

For each annulus the azimuthally averaged residual power spectrum is fitted over the adopted frequency interval $0.04 \leq k \leq 0.25$ using

$$P(k) = P_0 E(k) + P_1.$$

Here $E(k)$ is the expectation power spectrum obtained from the PSF convolved with the annular mask/window, P_0 is the fitted amplitude of the PSF-shaped component, and P_1 is a constant white-noise term. The fluctuation power attributed to the stellar population is

$$P_{\text{SBF}} = P_0 - P_r,$$

where P_r is the estimated contribution of unresolved globular clusters and background galaxies below the masking threshold. The apparent fluctuation magnitude follows from the photometric zero point and the mean galaxy model level in the annulus.

All current annular fits return negative P_1 . Because this behavior is nearly universal rather than confined to individual galaxies, it is treated as evidence that the constant-noise component is an imperfect description of the correlated noise in the drizzled images, not as an automatic rejection of all P_0 estimates. A final paper must quantify the sensitivity to the adopted frequency window and to a noise-template extension of the model.

3.3 Matched colors and uncertainties

F090W is PSF-matched and registered to F150W, and the integrated color is measured in exactly the same usable pixels and annuli as the SBF signal. This implements the central warning in Paper III that colors measured in the high-signal galaxy core can be biased relative to the regions used for SBF. The two ring measurements are retained separately and are also combined with inverse-variance weights.

The present internal error includes the fitting uncertainty, stability across the adopted k_{min} choices, the difference between the two annuli, PSF scatter, and a working uncertainty in P_r . Foreground-extinction, large-scale sky, and common TRGB zero-point uncertainties are not yet complete; the quoted error bars must therefore not be described as the final total error budget.

4 Results

4.1 F150W calibration

Using all 14 individual Paper IV TRGB distances, the preliminary linear calibration at the median color $C_0 = 0.5907$ is

$$\overline{M}_{150} = (-3.1726) + (0.6656)(C_{090-150} - 0.5907), \quad (1)$$

with an intrinsic scatter of 0.0695 mag. Bootstrap 16th–84th percentiles are $[-3.193, -3.147]$ mag for the intercept and $[0.190, 1.089]$ for the slope. The broad slope interval is a direct consequence of the narrow color range and the small sample.

A constant absolute SBF magnitude has $\text{AICc} = -22.83$, compared with -21.11 for the linear relation. The difference $\Delta\text{AICc} = 1.71$ is not decisive, while leave-one-out RMS values are 0.0984 and

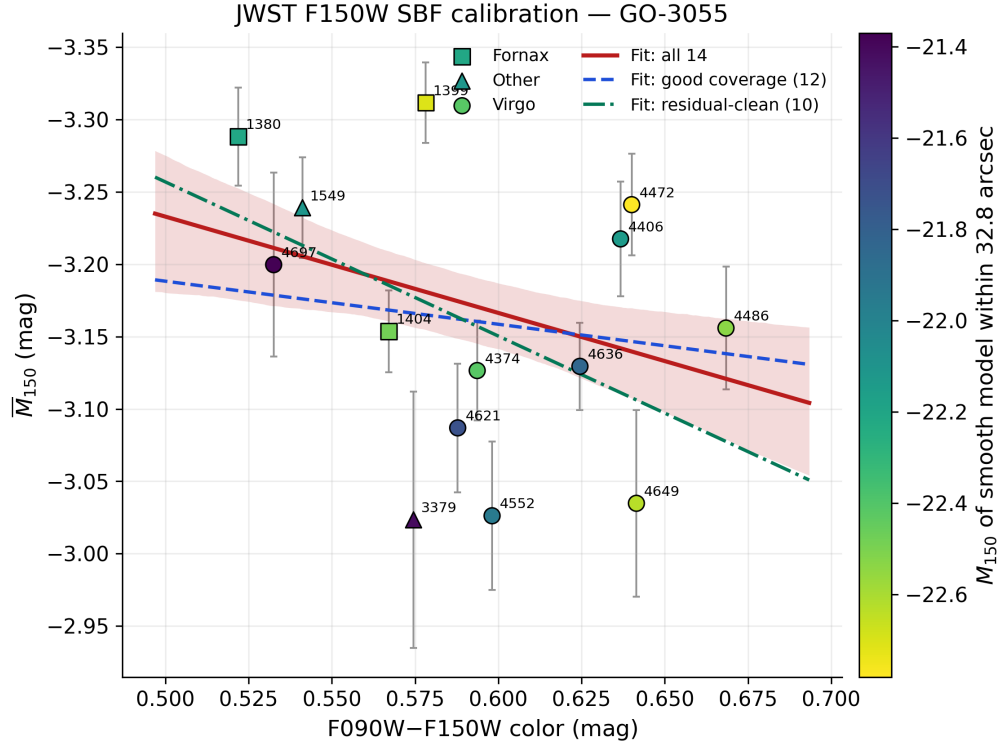


Figure 1: Direct JWST F150W SBF calibration for GO-3055. This is the JWST counterpart of the HST F110W calibration shown in Fig. 2 of Paper III. The vertical axis is inverted so that brighter SBF is upward. Before submission the secondary “clean” fit must use the residual-structure flag rather than radial annulus disagreement.

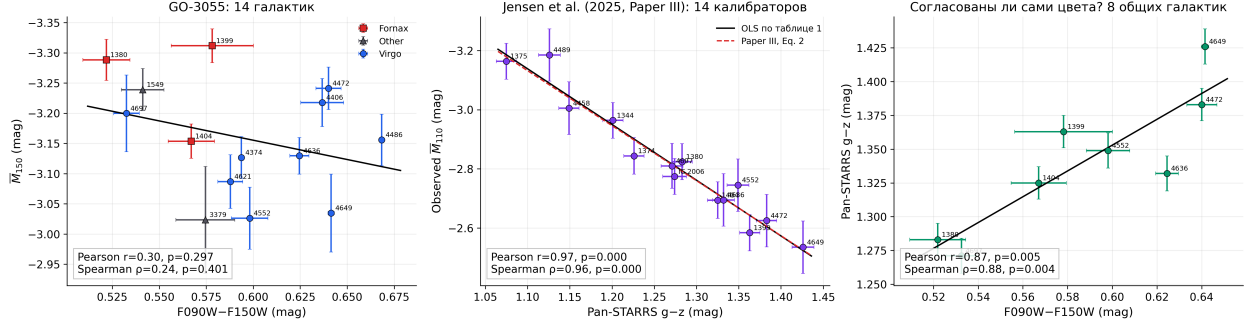


Figure 2: Observed SBF-color correlations in the present GO-3055 sample and in the 14 F110W calibration galaxies from Table 1 of Paper III, followed by a direct comparison of F090W-F150W and $g-z$ for the eight common galaxies. The filters and color indices differ, so slopes are not numerically interchangeable. The purpose is to compare the strength and robustness of the observed correlations without building the published color law into the dependent variable.

0.0987 mag respectively. Quadratic and broken-line fits reduce the raw RMS slightly but are disfavored after accounting for model complexity. We therefore report the linear calibration for direct comparison with Paper III but do not claim a secure detection of its color slope.

4.2 Observed color correlations

Paper III found a well-constrained F110W slope after enlarging the calibration set with cluster-mean distances and a broader color baseline. We compare unweighted Pearson and Spearman coefficients for our 14 objects, our measurements of the eight objects overlapping Table 1 of Paper III, and all 14 Paper III calibration points. For Paper III we compute $\bar{M}_{110}^{\text{obs}} = \bar{m}_{110} - \mu_{\text{cluster}}$ rather than correlating color with a magnitude already produced by their published color law. For our full sample, the unweighted coefficients are $r = 0.30$ ($p = 0.297$) and $\rho = 0.24$ ($p = 0.401$). Restricting our data to the eight Paper-III-overlap galaxies gives $r = 0.49$ ($p = 0.220$), whereas the 14 observed Paper III calibration points give $r = 0.97$ ($p < 10^{-8}$) and $\rho = 0.96$. Thus the current F150W data do not show a statistically significant raw color correlation, while the Paper III F110W relation is strong over its broader color baseline and cluster-augmented sample. The two color indices themselves are nevertheless strongly correlated for the eight common galaxies: F090W-F150W versus Pan-STARRS $g-z$ gives $r = 0.87$ ($p = 0.005$) and $\rho = 0.88$ ($p = 0.004$). The weak F150W SBF-color relation therefore cannot be dismissed as a complete failure of our infrared color to track the optical stellar-population sequence.

4.3 Radial measurements

The inner annuli give a slope of approximately 1.36 with intrinsic scatter 0.066 mag, whereas the outer annuli give a much weaker slope of approximately 0.28 and a larger scatter of 0.091 mag. The two measurements for a galaxy are correlated and must not be treated as 28 independent calibrators. Their difference is nevertheless a useful diagnostic of stellar-population gradients and residual modeling systematics.

NGC 3379 has the most spatially uniform normalized residual in the sample, but its annuli differ by 0.168 mag. The outer annulus is bluer by approximately 0.031 mag and has fainter SBF, in the opposite direction and at much larger amplitude than predicted by the global color relation. The unresolved-source correction is negligible in the inner annulus and below one per cent of P_0 in the outer annulus. NGC 3379

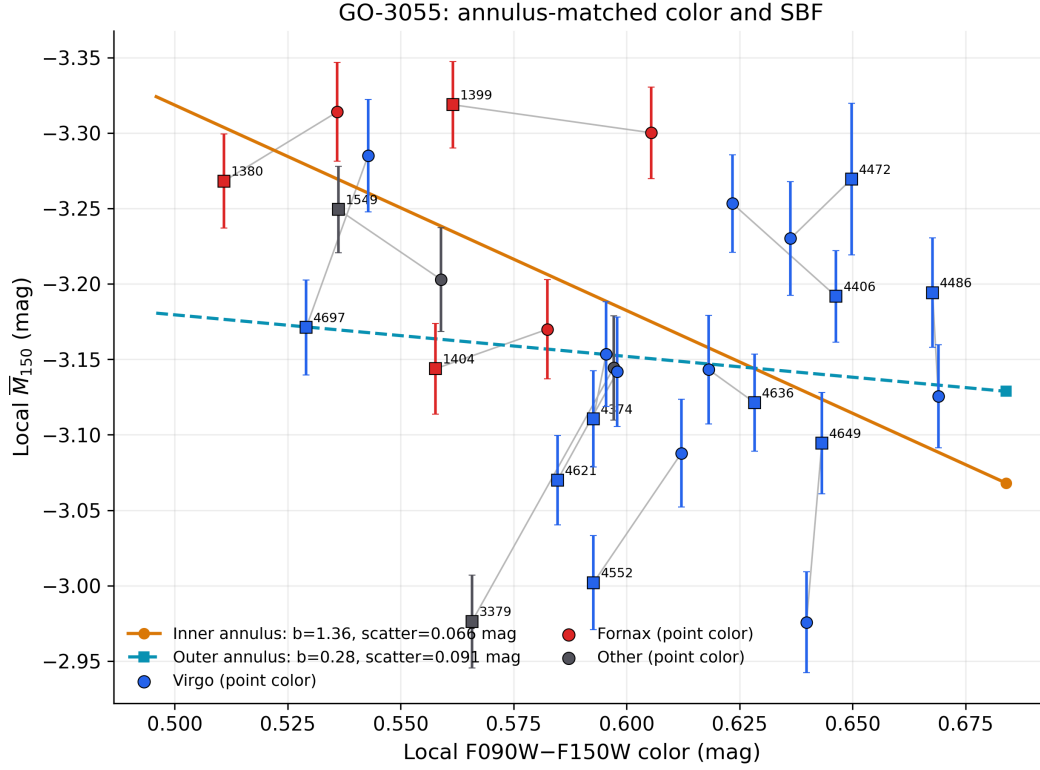


Figure 3: Annulus-matched local color and absolute F150W SBF magnitude. Circles and squares denote the inner and outer annuli, respectively; grey segments connect measurements of the same galaxy. Unlike Paper III, which reports the weighted annular means in its calibration table, we retain this ring-by-ring view to test whether the population correction is radially stable.

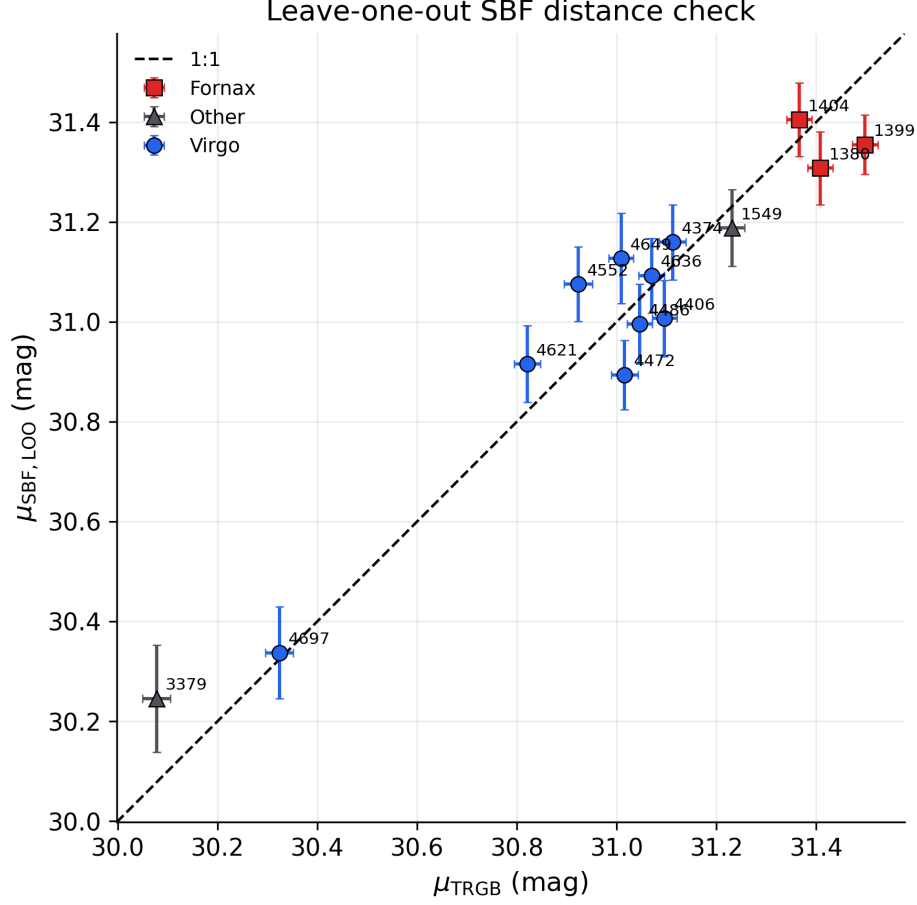


Figure 4: Leave-one-out F150W SBF distance moduli compared with the revised thin-slice TRGB moduli in Table 2 of Paper IV. This is the direct cross-validation of the JWST SBF rung against its resolved-star anchor rather than an in-sample reconstruction.

is therefore retained as a high-quality measurement and treated as evidence that annulus disagreement is not equivalent to poor residual structure.

4.4 Cross-validation against TRGB

For each galaxy, the color relation is refitted after excluding that galaxy, and the excluded SBF distance is compared with its Paper IV thin-slice TRGB distance. The mean residual is

$$\langle \mu_{\text{SBF, LOO}} - \mu_{\text{TRGB}} \rangle = +0.0077 \text{ mag},$$

with RMS 0.0987 mag and median absolute residual 0.0919 mag. The largest absolute residuals occur for NGC 3379, NGC 4552, NGC 1399, NGC 4472, and NGC 4649.

4.5 Dependence on host-galaxy luminosity

We test whether the F150W SBF magnitude correlates with the ordinary aperture luminosity of the host. The available quantity is the absolute F150W magnitude of the smooth galaxy model inside 32.8 arcsec, not a total galaxy magnitude. We therefore present both the direct correlation and the correlation of the residual about Eq. 1. The latter tests whether host luminosity carries information beyond the integrated

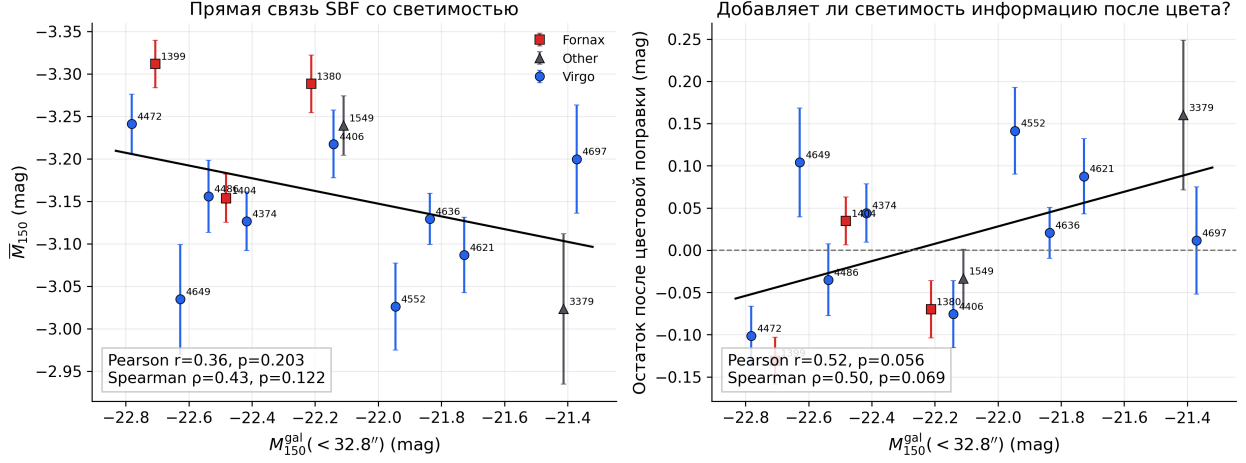


Figure 5: Absolute F150W SBF magnitude and color-corrected residual versus the absolute F150W magnitude of the smooth host model within 32.8 arcsec. A correlation in the left panel alone may reflect the usual color–luminosity relation of early-type galaxies; persistence in the right panel would indicate that luminosity adds independent predictive information.

Table 4: Numerical comparison of the published and preliminary calibrations. The slopes are not directly interchangeable because both the SBF band and the color index differ.

Quantity	Paper III	Present work
SBF band	HST F110W	JWST F150W
Color	PS $g - z$	F090W–F150W
Pivot color	1.30	0.5907
Slope	1.86 ± 0.16	$0.67^{+0.42}_{-0.48}$
Zero point	-2.760 ± 0.016	-3.173 ± 0.023
Calibration size	14	14
LOO RMS	not reported	0.099 mag

color. The direct Pearson coefficient is $r = 0.36$ ($p = 0.203$). After subtracting the linear color relation it increases to $r = 0.52$ ($p = 0.056$), with Spearman $\rho = 0.50$ ($p = 0.069$). This is not a formal detection, but it is sufficiently close to significance to motivate replacement of the fixed-aperture magnitude with homogeneous total photometry or stellar-mass estimates.

5 Comparison with the published calibrations

The present work is deliberately close to Paper III in measurement geometry: both use the same two annuli and their weighted mean, and both require the color to correspond to the SBF region. The comparison is not a literal repeat, because NIRCам F150W and F090W–F150W respond differently to stellar population age and metallicity than WFC3/IR F110W and optical $g - z$.

The distance-anchor treatment is also different. Paper III shows that using cluster means can reduce calibration scatter when individual measurement uncertainties are comparable to cluster depth. Our use of the individual Paper IV thin-slice distances is the stricter leave-one-galaxy-out test, but it also preserves line-of-sight depth and individual TRGB noise. Before submission, Eq. 1 must be recomputed

using the exact Paper III cluster-mean prescription. The individual-distance and cluster-mean results should be reported side by side rather than selecting the one with smaller scatter after seeing the answer.

Paper IV introduces a second controlled difference: its updated thin-slice moduli are on average 0.028 ± 0.008 mag below the earlier measurements. Consequently, an F150W zero point calibrated with Paper IV cannot be compared to the Paper III zero point without identifying the TRGB version. The anchor revision alone would shift an absolute SBF zero point at approximately the same 0.03 mag level, even if the measured apparent SBF magnitudes were unchanged.

6 Discussion

6.1 Is a color term detected?

The full-sample linear slope is positive and its bootstrap 16th percentile is above zero. This statement by itself is insufficient evidence for a physical law: a constant model has slightly better AICc and essentially identical leave-one-out performance. The fitted slope is also sensitive to the Virgo–Fornax balance, the radial annulus chosen, and the narrow color range. The correct conclusion is therefore that a Paper-III-like linear law is consistent with the data, but a color-independent F150W zero point is not excluded.

A tentative Virgo–Fornax offset of approximately -0.12 mag is visible. Although its bootstrap interval does not cross zero, adding the group step worsens AICc by approximately 0.98 because only three Fornax galaxies are available. It should be presented as an exploratory population/environment test, not as a detected second zero point.

6.2 Radial behavior and residual structure

The annular comparison is the most important diagnostic added beyond the Paper III presentation. A large annulus difference can occur in an exceptionally clean residual image, while a visually structured residual can have similar weighted SBF values in both annuli. Quality control must therefore separate model coverage, large-scale residual structure, and radial SBF gradients. NGC 4472, NGC 4486, NGC 4636, and NGC 4649 show the strongest large-scale residual structure in the current sector-based diagnostic. NGC 1380 and NGC 4697 have incomplete outer-annulus model coverage.

6.3 Current limitations

Three issues prevent the present draft from being submission-ready. First, the total systematic error budget must include foreground extinction, large-scale background uncertainty, the common NGC 4258/TRGB zero point, and the effect of the Paper IV thin/thick choice. Second, the universal negative P_1 values require a controlled correlated-noise test, ideally using a noise-template extension of the power-spectrum model. Third, the compact-source mask and P_r completeness model require injection or threshold sensitivity tests. These tests should be reported as changes in the final zero point and slope rather than as qualitative statements.

7 Conclusions

1. All 14 GO-3055 galaxies completed the F150W SBF measurement and artifact verification in the two Paper III annuli; all currently retain review-level quality flags, dominated by non-positive fitted P_1 .
2. The preliminary full-sample calibration has zero point -3.173 mag at $F090W - F150W = 0.5907$, slope 0.666, and intrinsic scatter 0.069 mag.

3. Leave-one-out comparison with the revised Paper IV TRGB distances gives an RMS of 0.099 mag, equivalent to approximately 4.7 per cent in distance, with negligible mean bias.
4. The present sample does not decisively distinguish a linear color correction from a constant F150W absolute fluctuation magnitude.
5. Inner and outer annuli do not follow the same empirical color trend. NGC 3379 demonstrates that a significant radial SBF difference can coexist with an exceptionally clean residual image.
6. A direct comparison with Paper III requires a second calibration using their cluster-mean distance prescription and an explicit accounting of the Paper IV TRGB zero-point revision.

A Distance comparison in Mpc

The complete machine-readable table is `runs/sbf2_go3055/analysis/tables/go3055_distances_mpc.csv`. It contains TRGB distances, leave-one-out F150W distances for each tested color law, and the available Jensen F160W values for the five overlapping galaxies. It is kept external to the main article table because including all four exploratory color laws obscures the adopted result.

B Reproducibility

The measurement notebook is `code/sbf-2.ipynb`; the dedicated GO-3055 runner is `code/run_sbf_2_batch.py`; and all analysis tables and figures are generated by `code/sbf-2-graph.ipynb`. The graph notebook reads the completed batch products and does not repeat the SBF measurement. Every figure is both saved as PNG/PDF and displayed inline for interactive inspection.

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